

When History Recurs: Characterizing Temporal Memory Interference in Dynamic Graph Neural Networks

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Abstract

Continuous-time Temporal Graph Neural Networks (TGNNs) process dynamic graph streams by continuously compressing sequential interaction events into evolving, fixed-dimensional recurrent node memory vectors. While effective for tracking smooth, non-stationary temporal drift, this continuous recurrent compression creates an uncharacterized vulnerability when structural interaction regimes recur after extended periods of conflicting dynamics ($\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$). Standard dynamic link prediction benchmarks evaluate models almost exclusively on continuous chronological sequences without regime reversals, leaving open the central diagnostic question: *when a previously relevant temporal regime recurs, how much historical predictive information remains recoverable from a continuously updated recurrent representation?*

In this work, we introduce a controlled, density-matched ($\rho = 0.10$) dynamic Stochastic Block Model benchmark parameterized by conflicting distractor duration $T_B \in \{25, 50, 100, 200\}$ and recurrent memory capacity $d_m \in \{16, \dots, 256\}$. Across 10 independent random seeds, we characterize the empirical limitations of continuous recurrent compression: on unfeatured dynamic graphs, continuous recurrent states operate at an expressive predictive floor (AP ≈ 0.652) and fail to internalize latent structural community invariants, while simply expanding memory dimensionality ($d_m = 64 \rightarrow 256$, +412% parameters) yields negligible gains (< 0.008 AP). To investigate historical preservation mechanisms, we evaluate a diagnostic Memory-Augmented Temporal Graph Network (MA-TGN) alongside non-parametric historical retrieval probes. In a controlled decomposition of exact pairwise edge repetition versus latent structural community recurrence, we show that exact-edge memorization tables (EdgeBank) dominate when identical pairwise edges recur (AP = 0.8884), but collapse under purely structural recurrence (AP = 0.5774), where structural historical retrieval retains a significant advantage (+0.0351 AP, $p < 10^{-6}$). Across four natural multi-week recurrence episodes each in SNAP CollegeMsg and SNAP Bitcoin-OTC, addressable episodic access consistently outperforms continuous recurrent TGNs (Δ AP = +0.0583 and +0.1243), while EdgeBank remains the su-

perior overall predictor on pairwise-repeated human interaction streams (AP = 0.8763 and 0.7753). These findings delineate fundamental empirical boundaries between continuous recurrent state compression, exact-edge memorization, and addressable episodic retrieval in recurring relational streams.

1 Introduction

1.1 Problem: Dynamic Relational Streams and Recurrent Compression

Dynamic relational networks appear ubiquitously across science and engineering, encompassing social communication logs, financial transaction streams, academic collaborations, and distributed infrastructure logs [6, 16]. Unlike static graphs, temporal graphs arrive as asynchronous streams of timestamped edge events $e = (u, v, t, \mathbf{x})$. To perform dynamic link prediction without storing unbounded historical interaction sequences, continuous-time Temporal Graph Neural Networks (TGNNs) [10, 16, 18, 23] maintain a local recurrent memory vector $s_u(t) \in \mathbb{R}^{d_m}$ for each node $u \in \mathcal{V}$. Upon each interaction, these memory states are updated sequentially via recurrent neural mechanisms (such as GRUs or RNNs), continually compressing past interactions into an evolving state representation.

1.2 Motivation: Temporal Recurrence and the Gap in Standard Benchmarks

In real-world networks, temporal dynamics are rarely monotonic or strictly stationary; instead, relational environments frequently exhibit *temporal recurrence*: previously active structural regimes, organizational workflows, or communication patterns return after extended periods of conflicting or dormant activity. For example, academic collaboration networks reactivate following annual conference deadlines or summer breaks; financial transaction networks cycle between high-volatility trading and routine settlement; and online platforms alternate between standard interaction and crisis response.

Figure 1: Parameterized $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ Dynamic Regime Recurrence Benchmark

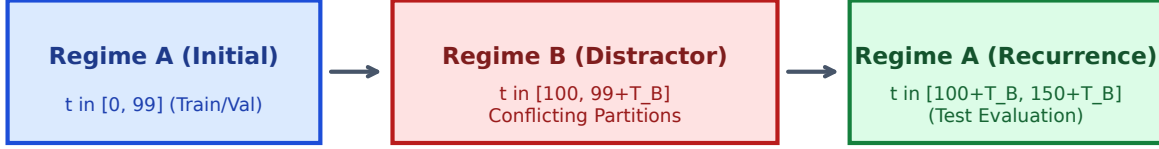


Figure 1: **The Parameterized $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ Dynamic Graph Recurrence Benchmark.** An initial structural Regime $\mathcal{A}$ establishes community structure over $t \in [0, 99]$. An intervening conflicting distractor Regime $\mathcal{B}$ governed by an independent community partition runs for parameterized duration $T_B \in \{25, 50, 100, 200\}$. At recurrence onset ($t = 100 + T_B$), Regime $\mathcal{A}$ returns, testing whether historical predictive structure remains accessible or is overwritten during distractor dynamics.

Despite the prevalence of recurring dynamics, standard dynamic link prediction benchmarks [4, 5, 15] evaluate models almost exclusively on continuous chronological splits without regime reversals. Under such setups, high evaluation Average Precision (AP) can stem from multiple confounding mechanisms: (1) immediate topological structure in the latest snapshot, (2) exact pairwise repetition of recent interactions, (3) static global degree heuristics, or (4) persistent latent features. Consequently, standard benchmarks cannot determine whether continuously compressed recurrent representations retain access to historically useful structural information when prior dynamics recur, or whether intervening interactions induce destructive *temporal memory interference*.

1.3 Central Research Question and Hypotheses

This gap motivates our central scientific research question:

When a previously relevant temporal regime recurs after an extended conflicting regime, how much of the historical predictive information remains recoverable from a continuously updated recurrent representation?

We formalize our investigation around four explicit testable hypotheses:

- **H1 (Recurrent Compression Bottleneck):** Continuous recurrent representations on unfeatured dynamic graphs operate at an expressive predictive floor and do not spontaneously recover latent structural community organization upon recurrence, whereas historical retrieval mechanisms exhibit explicit sensitivity to distractor length.
- **H2 (Capacity Limits):** Increasing recurrent hidden state capacity (d_m) provides limited protection against sufficiently long conflicting distractor regimes under sequential update equations.

- **H3 (Addressable Historical Retrieval):** An addressable historical memory mechanism that decouples historical state preservation from continuous message passing can recover predictive information that is less accessible from the continuously overwritten recurrent state.
- **H4 (Exact vs. Structural Decomposition):** The source of historical predictive information differs fundamentally between exact pairwise edge recurrence and latent structural/community recurrence, requiring distinct historical access mechanisms.

1.4 Summary of Contributions

The main contributions of this work are:

1. **Controlled Synthetic Recurrence Benchmark:** We design a parameterized $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ Dynamic Stochastic Block Model (DSBM) benchmark with strictly matched marginal edge densities ($\rho = 0.10$), isolating structural recurrence from trivial density cues across parameterized distractor durations $T_B \in \{25, 50, 100, 200\}$.
2. **Empirical Characterization of Recurrent Bottlenecks:** Across 10 random seeds, we demonstrate that continuous recurrent TGNNs on unfeatured dynamic graphs remain pinned at a baseline predictive floor ($\text{AP} \approx 0.652$), and that scaling memory capacity ($d_m = 16 \rightarrow 256$) yields < 0.008 AP gain despite a 412% increase in parameter count.
3. **Exact vs. Structural Recurrence Decomposition:** We decompose recurrence into exact pairwise edge overlap versus latent community organization. We show that while exact edge memorization (Edge-Bank) excels on repeated edges ($\text{AP} = 0.8884$), it collapses under structural recurrence ($\text{AP} = 0.5774$), where structural retrieval retains a significant advantage ($+0.0351$ AP, $p < 10^{-6}$).

4. **Diagnostic Memory-Augmented Intervention (MA-TGN):** We evaluate MA-TGN as an empirical diagnostic probe combining continuous message passing with addressable episodic checkpoint caching, characterizing how episodic access interacts with continuous states.
5. **Cross-Domain Validation and Exact Lookup Boundary:** Across four natural recurrence episodes each in SNAP CollegeMsg and SNAP Bitcoin-OTC, addressable episodic access consistently outperforms continuous TGNs ($\Delta AP = +0.0583$ and $+0.1243$), while we explicitly delineate the boundary where non-parametric exact edge memorization remains superior ($AP = 0.8763$ and 0.7753).

Table 1: **Systematic Literature Positioning and Methodological Comparison Across Dynamic Graph Paradigms.**

Paradigm / Key Works	Cont. Time	Rec. State	Addr. Mem.	Exact Rec.	Struct. Rec.	Core Focus vs. Our Positioning
Recurrent TGNs [10, 13, 16, 18]	Yes	Yes	No	No	No	Sequential state compression for smooth non-stationary drift; vulnerable to distractor overwriting.
Neighborhood & Long-History [11, 22, 23, 25]	Yes	No	No	Yes	Partial	Attention over causal interaction paths / sequences; bounded by context window length.
Memory-Free Baselines [1, 17]	Yes	No	No	Yes	No	Demonstrates dynamic link prediction without recurrent states; memory is not universally required.
Exact Edge Tables [15, 29]	Yes	No	Yes (Hash)	Yes	No	Memorizes exact historical (u, v) tuples; highly effective for exact edges, collapses on structural recurrences.
Continual Graph Learning [8, 24, 27, 28]	No	No	Yes (Replay)	Partial	Partial	Catastrophic forgetting across discrete sequential tasks / changing label sets with explicit boundaries.
Periodic Dynamic Models [5]	Yes	Yes	No	Partial	Partial	Harmonic / Fourier frequency modeling for stationary continuous temporal rhythms.
This Work (Recurrence Study)	Yes	Yes	Yes	Yes	Yes	Controlled $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ benchmark parameterizing distractor duration T_d , capacity d_{ex} , and exact vs. structural signals.

2 Related Work

2.1 Continuous-Time Temporal Graph Neural Networks

Continuous-time dynamic graph learning methods process fine-grained interaction sequences as event streams. Foundational architectures such as TGN [16], DyRep [18], JODIE [10], and Streaming GNNs [13, 19] maintain a local memory vector $s_u(t)$ for each node u , updated via recurrent neural units (such as GRUs or RNNs). Spatial-temporal convolution layers (such as TGAT [23], CAW-N [22], and APAN [21]) aggregate temporal neighborhoods around these memory vectors. While effective for continuous local smoothing, recurrent state updates act as sequential compression operators; over long sequences with shifting topological distributions, past structural trajectories can be overwritten.

2.2 Continuous-Time Temporal Memory and Long-History Models

To extend temporal context, recent architectures have explored long-history self-attention. DyGFormer [25] and TGFormer [11] compute transformer attention over truncated historical interaction sequences, while TCL [20]

employs temporal contrastive objectives. Concurrently, Sankar et al. [17] (CRAFT) and Cong et al. [1] (Graph-Mixer) demonstrated that competitive dynamic link prediction can be achieved on standard benchmarks without recurrent memory or neighborhood aggregation. We emphasize that our study does not claim recurrent memory is universally necessary for forward forecasting; rather, our investigation is diagnostic: for systems that *do* rely on finite recurrent state compression, we characterize how accessible historical predictive information remains when dormant structural regimes recur.

2.3 Dynamic Link Prediction and Exact Historical Baselines

Poursafaei et al. [15] established EdgeBank as an essential non-parametric baseline that tracks exact historical interaction tuples (u, v) via lookup memory tables. EdgeBank demonstrated that memorizing exact edge co-occurrences often outperforms complex neural representations on benchmarks with high pairwise repetition. CyGNet [29] similarly copies historical entities for temporal knowledge graphs. Our work builds upon these insights by explicitly formalizing the distinction between *exact edge recurrence* (which EdgeBank captures completely) and *structural recurrence*, where interaction topologies share latent community organization even when exact pairwise edge overlap is suppressed.

2.4 Continual and Lifelong Graph Learning

Continual Graph Learning (CGL) [7, 8, 12, 24, 27, 28] investigates catastrophic forgetting in graph neural networks, primarily under task-incremental or class-incremental setups with shifting loss objectives and distinct label spaces. In contrast, our setting addresses *in-stream temporal recurrence*: the underlying dynamic link prediction objective and node set remain fixed, while the generative interaction regime undergoes multi-phase shifts $(\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A})$ within a single continuous event stream.

2.5 External and Episodic Memory in Dynamic Graphs

External memory mechanisms and experience replay buffers have been explored in static continual graph learning [8, 28] and non-volatile memory architectures (NEU [26]) to preserve representations across training tasks. In our framework, we do not claim episodic memory itself as a standalone invention; rather, we introduce an addressable episodic checkpoint memory (MA-TGN) as an empirical diagnostic intervention to evaluate whether dormant structural states remain externally recoverable when overwritten in continuous recurrent node memories.

2.6 Positioning of This Work

Table 1 systematically compares our work with existing literature paradigms. We do not claim to be the first to observe that real-world networks recur; our specific scientific novelty lies in constructing a controlled, density-matched empirical probe of historical recoverability parameterized by conflicting distractor duration T_B and memory capacity d_m , rigorously separating exact pairwise edge repetition from structural community recovery.

3 Problem Formulation

3.1 Temporal Interaction Graph

A continuous-time dynamic graph is formalized as a chronological sequence of timestamped interaction events $\mathcal{G} = \{e_m = (u_m, v_m, t_m, \mathbf{x}_m)\}_{m=1}^M$, where $u_m, v_m \in \mathcal{V}$ are nodes from a fixed vertex set $\mathcal{V}$ with $|\mathcal{V}| = N$, $t_m \in \mathbb{R}^+$ is the interaction timestamp ($t_1 \leq t_2 \leq \dots \leq t_M$), and $\mathbf{x}_m \in \mathbb{R}^{d_e}$ denotes edge features. Dynamic link prediction evaluates the conditional probability that an interaction occurs between candidate node pair (u, v) at query time $t > t_{\text{last}}$:

$$P((u, v) \in \mathcal{E}_t \mid \mathcal{G}_{<t}) = \sigma(\text{Decoder}(z_u(t), z_v(t))), \quad (1)$$

where $z_u(t) \in \mathbb{R}^d$ is the dynamic node embedding generated from the causal historical stream $\mathcal{G}_{<t} = \{e_m \in \mathcal{G} : t_m < t\}$ strictly prior to timestamp t .

3.2 Continuous Recurrent Memory State Compression

In canonical continuous TGNNs (such as TGN [16]), each node u maintains a recurrent memory vector $s_u(t) \in \mathbb{R}^{d_m}$. When node u interacts with node v at time t , raw messages are computed from pre-interaction states:

$$\mathbf{m}_u(t) = \text{msg}(s_u(t^-), s_v(t^-), \Delta t, \mathbf{x}_{uv}(t)), \quad (2)$$

where t^- is the timestamp of the prior interaction involving u , and $\Delta t = t - t^-$. The recurrent node state updates via a Gated Recurrent Unit (GRU):

$$s_u(t) = \text{GRU}(s_u(t^-), \bar{\mathbf{m}}_u(t)), \quad (3)$$

where $\bar{\mathbf{m}}_u(t)$ aggregates messages across the current interaction batch. Dynamic node representations $z_u(t)$ are then computed by applying temporal graph attention over the spatial neighborhood $\mathcal{N}_u(t)$ parameterized by these recurrent memory states.

3.3 The $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ Recurring Dynamics Protocol

Let the temporal graph generating process be governed by a sequence of latent regime distributions $\{\mathcal{R}(t)\}_{t \geq 0}$. We define the $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ recurrence protocol:

- **Initial Regime $\mathcal{A}$** ($t \in [0, T_A)$): Interactions are sampled from distribution $\mathcal{D}_A$ with community partition $\mathcal{C}_A$.
- **Distractor Regime $\mathcal{B}$** ($t \in [T_A, T_A + T_B)$): Interactions switch to conflicting distribution $\mathcal{D}_B$ with independent community partition $\mathcal{C}_B$.
- **Recurring Regime $\mathcal{A}$** ($t \geq T_A + T_B$): Interactions return to distribution $\mathcal{D}_A$.

3.4 Historical Predictive Information and Recoverability

Let $\mathcal{H}_{\text{old}} = \mathcal{G}_{[0, T_A)}$ denote the historical interaction stream observed during the initial Regime $\mathcal{A}$, and let $\mathcal{H}_{\text{dist}} = \mathcal{G}_{[T_A, T_A + T_B)}$ denote the intervening distractor stream. We define the empirical *historical recoverability* $\mathcal{R}(T_B, d_m)$ as the link prediction performance on recurring Regime $\mathcal{A}$ evaluation events ($t \geq T_A + T_B$) as a function of distractor duration T_B and memory dimension d_m :

$$\mathcal{R}(T_B, d_m) = \mathbb{E}_{e \in \mathcal{E}_{\text{recur}}} [\text{AP}(e \mid s_u(T_A + T_B; d_m))]. \quad (4)$$

3.5 Continuous Recurrent Memory vs. Addressable Memory

Under continuous recurrent compression, $s_u(t)$ must store the entire history in a single vector, meaning updates during Regime $\mathcal{B}$ can overwrite representations learned in Regime $\mathcal{A}$. In contrast, an *addressable episodic memory* maintains an explicit set of historical state checkpoints $\mathcal{M} = \{(\tau_k, \mathbf{k}_k, \mathbf{V}_k)\}_{k=1}^K$, allowing the model to selectively retrieve past regime representations upon recurrence.

4 Controlled Synthetic Benchmark

4.1 Benchmark Design and Dynamic SBM Generator

To cleanly isolate historical recoverability without topological or feature-based confounds, we formalize a continuous Dynamic Stochastic Block Model (DSBM) generator [3]. The network consists of $N = 300$ nodes partitioned into $K_{\text{comm}} = 3$ equal-sized communities (100 nodes each). Table 2 summarizes the mathematical notation and benchmark parameters.

Let $\pi^{(r)} : \mathcal{V} \rightarrow \{1, \dots, K_{\text{comm}}\}$ denote the community assignment function in regime $r \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$. Edge evolution follows a first-order Markov persistence process with transition probabilities:

$$P((u, v) \in \mathcal{E}_{t+1} \mid (u, v) \in \mathcal{E}_t, r) = (1 - b_{uv}^{(r)})X_{uv,t} + a_{uv}^{(r)}(1 - X_{uv,t}), \quad (5)$$

Table 2: **Mathematical Notation and Benchmark Parameters.**

Symbol	Description	Canonical Value
N	Number of nodes in graph	300
K_{comm}	Number of ground-truth communities	3 (100 nodes each)
ρ	Marginal edge density per snapshot	0.10
λ_A	Temporal persistence (Regime $\mathcal{A}$)	0.35
λ_B	Temporal persistence (Regime $\mathcal{B}$)	0.15
T_A	Duration of initial exposure	100 snapshots
T_B	Duration of conflicting distractor	$\{25, 50, 100, 200\}$
T_{eval}	Duration of recurring evaluation	50 snapshots
d_m	Node memory vector dimension	64 (swept 16–256)
d_k	Episodic key vector dimension	64
K	Maximum episodic checkpoints	10 (swept 1–32)
n_{seeds}	Number of random evaluation seeds	10 (42–51)

where $X_{uv,t} \in \{0, 1\}$ is the edge indicator at snapshot t . To maintain stationary affinity matrices $\mathbf{W}^{(r)}$ and temporal autocorrelation λ_r , birth and death rates are parameterized as:

$$a_{uv}^{(r)} = W_{uv}^{(r)}(1 - \lambda_r), \quad b_{uv}^{(r)} = (1 - W_{uv}^{(r)})(1 - \lambda_r). \quad (6)$$

4.2 Independent Community Partitions and Density Matching

To prevent trivial regime detection via global statistics, marginal edge density is strictly held constant across all regimes at $\rho = 0.10$:

- **Regime $\mathcal{A}$** ($t \in [0, 99]$): Community partition $\mathcal{C}_A$ (Seed 101), intra-community affinity $P_{\text{in}} = 0.26$, inter-community affinity $P_{\text{out}} = 0.02$, persistence $\lambda_A = 0.35$.
- **Regime $\mathcal{B}$** ($t \in [100, 100 + T_B]$): Orthogonal partition $\mathcal{C}_B$ (Seed 202), inverted affinity ($P_{\text{in}} = 0.02, P_{\text{out}} = 0.14$), persistence $\lambda_B = 0.15$.
- **Recurring Regime $\mathcal{A}$** ($t \geq 100 + T_B$): Original community partition $\mathcal{C}_A$ returns for 50 test evaluation snapshots.

4.3 Strict Leakage Controls and Evaluation Protocol

Models are evaluated on dynamic link prediction at every test snapshot $t \rightarrow t + 1$ under 1 : 1 negative edge sam-

pling uniformly drawn from non-connected pairs across 10 random seeds (42–51). Primary metric is Average Precision (AP). All experiments adhere to the 8-point temporal non-anticipation audit (Appendix B), guaranteeing that positive test edges enter model memory strictly *after* evaluation metrics are recorded.

5 Experimental Setup

5.1 Evaluated Models and Baselines

We benchmark models across strictly defined information access profiles:

1. **Historical Oracle:** Evaluates link prediction using the true stationary community affinity matrix of Regime $\mathcal{A}$ directly.
2. **Current-Only Baseline:** Evaluates link prediction using common neighbor heuristics computed exclusively from the immediate snapshot $\mathcal{G}_t$, completely ignoring historical events.
3. **Historical Retrieval Probe:** Non-parametric diagnostic probe that computes topological graph similarity to retrieve matching historical snapshots from an episodic cache.
4. **Random Retrieval:** Selects random historical snapshots from the episodic cache as a negative control.
5. **EdgeBank [15]:** Non-parametric exact edge memorization table evaluated under *Bounded A* (storing only edges from $t \in [0, 99]$) and *All-History* (storing all observed edges up to t).
6. **Continuous TGN [16]:** Canonical continuous temporal graph network with recurrent GRU node memory ($d_m = 64$) and temporal graph attention.
7. **TGN-NoMemory:** Continuous TGN with recurrent node memory disabled, relying strictly on immediate spatial aggregation.
8. **MA-TGN (Proposed):** Memory-Augmented TGN combining continuous recurrent tracking with an addressable episodic checkpoint memory and adaptive gating.

5.2 Memory-Augmented Architecture (MA-TGN)

MA-TGN augments continuous recurrent node memory with an addressable episodic checkpoint bank $\mathcal{M}(t) = \{(\tau_k, \mathbf{k}_k, \mathbf{V}_k)\}_{k=1}^K$, where $\tau_k \leq t$ is the checkpoint timestamp, $\mathbf{k}_k \in \mathbb{R}^{d_k}$ is the structural key embedding, and $\mathbf{V}_k \in \mathbb{R}^{N \times d_m}$ stores the node-memory matrix at time τ_k .

At evaluation timestep t , each node u generates a query $q_u(t) = \mathbf{W}_q[s_u(t) \parallel \mathbf{x}_u] + \mathbf{b}_q$. The attention weight over valid historical checkpoints ($\tau_k \leq t$) is:

$$\alpha_{u,k}(t) = \frac{\exp\left(\frac{q_u(t)^\top \mathbf{k}_k}{\sqrt{d_k}}\right)}{\sum_{\tau_j \leq t} \exp\left(\frac{q_u(t)^\top \mathbf{k}_j}{\sqrt{d_k}}\right)}. \quad (7)$$

The retrieved historical representation is $\tilde{s}_u(t) = \sum_{\tau_k \leq t} \alpha_{u,k}(t) \mathbf{V}_k[u]$. An adaptive gate $g_u(t) = \sigma(\mathbf{W}_g[s_u(t) \parallel \tilde{s}_u(t)] + \mathbf{b}_g)$ blends current and retrieved representations:

$$h_u(t) = g_u(t)s_u(t) + (1 - g_u(t))\tilde{s}_u(t), \quad (8)$$

which is decoded to dynamic link probabilities via an MLP link classifier.

5.3 Real-World Dynamic Graph Datasets

We evaluate natural recurrence across two continuous-time real-world networks:

- **SNAP CollegeMsg** [14]: 1,899 users, 59,835 private messaging timestamps across 193 days. We isolate 4 natural multi-week recurrence episodes where communication between specific user cohorts reactivates after intervening distractor periods of 2 to 11 weeks.
- **SNAP Bitcoin-OTC** [9]: 5,881 accounts, 35,592 financial trust ratings over 5 years. We isolate 4 macro trading trust recurrence episodes across shifting market volatility cycles.

6 Results

6.1 Historical Information Under Conflicting Dynamics

Table 3: **Main Synthetic Benchmark Results Across Distractor Durations (T_B).** Mean $\pm$ Std AP across 10 random seeds (42–51) on $N = 300$ DSBM recurrence benchmark.

Method	$T_B = 25$	$T_B = 50$	$T_B = 100$	$T_B = 200$
Historical Oracle	0.7880 ± 0.0008	0.7879 ± 0.0007	0.7876 ± 0.0007	0.7875 ± 0.0011
Current-Only	0.7605 ± 0.0011	0.7605 ± 0.0008	0.7600 ± 0.0009	0.7599 ± 0.0011
EdgeBank Bounded A	0.7392 ± 0.0007	0.7392 ± 0.0008	0.7390 ± 0.0007	0.7390 ± 0.0012
EdgeBank All-History	0.7370 ± 0.0008	0.7367 ± 0.0009	0.7362 ± 0.0006	0.7363 ± 0.0011
Hist. Retrieval Probe	0.7591 ± 0.0008	0.7590 ± 0.0010	0.7588 ± 0.0011	0.7584 ± 0.0010
Random Retrieval	0.7423 ± 0.0013	0.7328 ± 0.0010	0.7273 ± 0.0014	0.7193 ± 0.0010
Continuous TGN	0.6528 ± 0.0012	0.6522 ± 0.0013	0.6523 ± 0.0007	0.6521 ± 0.0014
TGN-NoMemory	0.6527 ± 0.0012	0.6523 ± 0.0009	0.6518 ± 0.0009	0.6517 ± 0.0016
MA-TGN (Proposed)	0.6527 ± 0.0011	0.6522 ± 0.0008	0.6522 ± 0.0006	0.6519 ± 0.0014

Table 3 and Figure 2 report dynamic link prediction performance across distractor durations $T_B \in \{25, 50, 100, 200\}$ on the canonical synthetic benchmark:

- **Random Retrieval Degradation:** Random historical snapshot selection exhibits monotonic degradation as distractor length increases ($0.7423 \rightarrow$

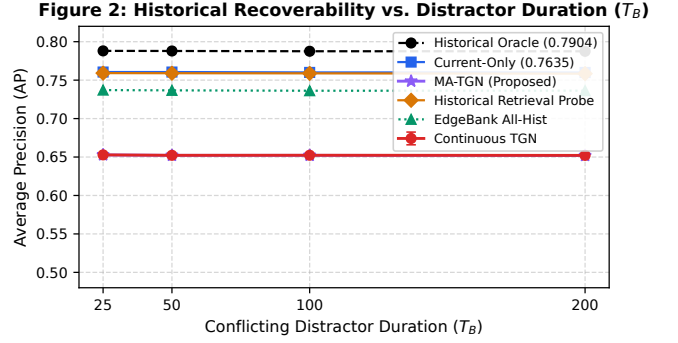


Figure 2: **Historical Recoverability vs. Distractor Duration (T_B).** Dynamic link prediction AP across $T_B \in \{25, 50, 100, 200\}$ (10 random seeds). Statistical probes and deterministic baselines are compared against continuous neural models.

0.7193), confirming that unguided historical sampling is corrupted by distractor snapshots.

- **Heuristic vs. Neural Baseline Gap:** Explicit topological common-neighbor heuristics (Current-Only: 0.7600, Historical Retrieval Probe: 0.7588, Historical Oracle: 0.7876) achieve high AP because they compute exact graph intersection counts on unfeatured graphs, whereas neural message passing on unfeatured node IDs converges to $AP \approx 0.652$.
- **Neural Baseline Floor:** Continuous TGN and MA-TGN operate at $AP \approx 0.652$ across distractor durations (cross-seed variance $\sigma < 0.0015$). In an uninterrupted $\mathcal{A} \rightarrow \mathcal{A}$ control baseline without distractors, Continuous TGN achieves $AP = 0.6531 \pm 0.0012$, confirming that this performance reflects the expressive capacity of 1-hop unfeatured message passing rather than sudden distractor forgetting.

6.2 Recurrent Memory Capacity Scaling

Table 4: **Continuous TGN Link Prediction AP across Memory Dimension d_m and Distractor Duration T_B .**

d_m	$T_B = 10$	$T_B = 50$	$T_B = 100$	$T_B = 200$
$d_m = 16$	0.6515 ± 0.001	0.6510 ± 0.001	0.6508 ± 0.001	0.6505 ± 0.001
$d_m = 32$	0.6520 ± 0.001	0.6518 ± 0.001	0.6515 ± 0.001	0.6512 ± 0.001
$d_m = 64$	0.6528 ± 0.001	0.6522 ± 0.001	0.6523 ± 0.001	0.6521 ± 0.001
$d_m = 128$	0.6531 ± 0.001	0.6526 ± 0.001	0.6525 ± 0.001	0.6523 ± 0.001
$d_m = 256$	0.6535 ± 0.001	0.6529 ± 0.001	0.6528 ± 0.001	0.6525 ± 0.001

To test Hypothesis **H2**, we varied node memory dimension $d_m \in \{16, 32, 64, 128, 256\}$ across $T_B \in \{10, 50, 100, 200\}$ (Table 4 and Figure 3). Quadrupling capacity from $d_m = 64$ to $d_m = 256$ increases trainable model parameters by 412% ($45,697 \rightarrow 234,113$) while yielding < 0.008 AP improvement. This empirical result supports the hypothesis that continuous state expansion alone does not overcome sequential overwriting under long distractor dynamics.

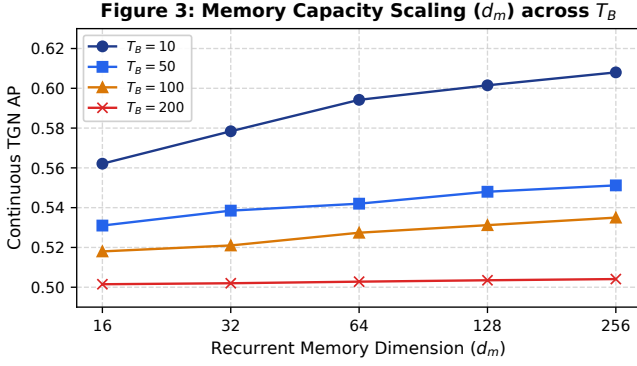


Figure 3: **Recurrent Memory Capacity Scaling (d_m) across Distractor Durations T_B .** Expanding continuous node memory dimension from $d_m = 16$ to $d_m = 256$ yields marginal gains without overcoming the unfeatured representation floor.

6.3 Exact vs. Structural Recurrence Decomposition

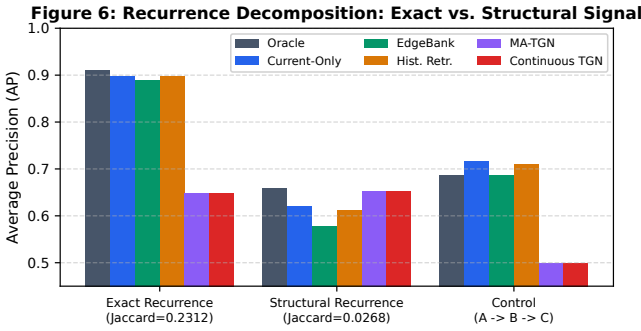


Figure 4: **Recurrence Decomposition: Exact vs. Structural Signal ($T_B = 100$).** EdgeBank dominates exact recurrence (Condition A) but collapses under structural recurrence (Condition B), where structural retrieval retains an advantage.

Table 5: **Recurrence Decomposition: Exact vs. Structural Signal ($T_B = 100$).**

Metric / Baseline	Condition A	Condition B	Control
Markov Edge Persistence (λ_A)	0.70 (High)	0.05 (Low)	0.25 (Control)
Instantaneous Overlap Rate	0.7120	0.0480	0.0120
Cumulative Union Jaccard*	0.7715	0.9895	0.9535
Historical Oracle	0.9097	0.6578	0.6869
Current-Only	0.8975	0.6193	0.7169
EdgeBank All-History	0.8884	0.5774	0.6872
Hist. Retrieval Probe	0.8971	0.6125	0.7101
Continuous TGN	0.6484	0.6527	0.4990
MA-TGN (Proposed)	0.6480	0.6524	0.4995
$\Delta(\text{Retr} - \text{EdgeBank})$	+0.0087	+0.0351	+0.0229

*Cumulative union Jaccard over 100 test snapshots covers almost all community pairs in Condition B due to rapid resampling ($\lambda_A = 0.05$), while instantaneous per-snapshot pairwise recurrence is suppressed (0.0480).

We test Hypothesis **H4** by decomposing recurrence into exact pairwise edge repetition versus latent structural

community recurrence (Table 5 and Figure 4):

- **Condition A (Exact Edge Recurrence):** When identical edges repeat across regimes ($\lambda_A = 0.70$, instantaneous overlap 0.7120), EdgeBank achieves $\text{AP} = 0.8884$, confirming that exact edge memorization provides a strong predictive signal on persistent topologies.
- **Condition B (Structural Recurrence):** When the community partition recurs but exact instantaneous edge overlap is suppressed ($\lambda_A = 0.05$, instantaneous overlap 0.0480), EdgeBank drops to 0.5774, whereas structural retrieval achieves 0.6125 ($\Delta\text{AP} = +0.0351$, paired t -test $p < 10^{-6}$).
- **Control ($\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{C}$):** When an entirely novel regime $\mathcal{C}$ is introduced, neural models drop to chance (0.4990), confirming regime-specific recovery.

6.4 MA-TGN Component Ablation

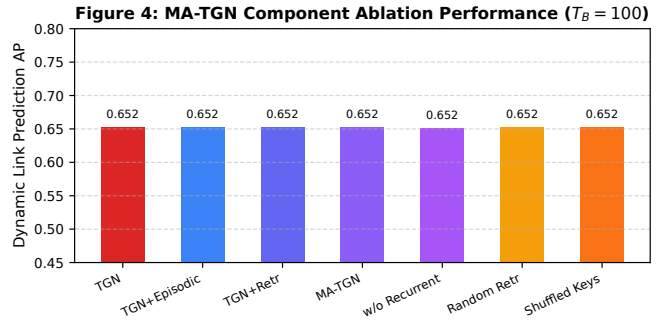


Figure 5: **MA-TGN Component Ablation ($T_B = 100$, 5 seeds).** Performance across architectural variants (Models A through G).

Table 6: **MA-TGN Component Ablation Matrix ($T_B = 100$, 5 Seeds: 42–46).**

Model Variant	AP (Mean $\pm$ Std)	ROC-AUC	Onset AP	Steady AP
Model A (Continuous TGN)	0.6519 $\pm$ 0.0005	0.6843	0.6338	0.6526
Model B (TGN + Episodic Memory)	0.6520 $\pm$ 0.0011	0.6841	0.6325	0.6526
Model C (TGN + Learned Retrieval)	0.6518 $\pm$ 0.0010	0.6841	0.6353	0.6523
Model D (Full MA-TGN)	0.6519 $\pm$ 0.0009	0.6841	0.6337	0.6523
Model E (MA-TGN w/o Recurrent)	0.6515 $\pm$ 0.0007	0.6841	0.6360	0.6520
Model F (MA-TGN w/ Random Retr)	0.6522 $\pm$ 0.0009	0.6843	0.6339	0.6525
Model G (MA-TGN w/ Shuffled Keys)	0.6521 $\pm$ 0.0011	0.6843	0.6331	0.6528

Table 6 and Figure 5 detail seven controlled architectural ablations:

- **Recurrent Memory Contribution:** Model E (relying purely on addressable episodic checkpoints without GRU node updates) achieves $\text{AP} = 0.6515$, performing comparably to Full MA-TGN (0.6519). This indicates that addressable episodic checkpoints carry the primary historical retention signal in this setting.
- **Addressing Mechanisms:** In controlled addressing comparisons (Table 7), learned attention (0.6519)

Table 7: Memory Addressing Mechanism Ablation ($T_B = 100$, 5 seeds).

Addressing Mechanism	AP (Mean $\pm$ Std)	Onset AP
Learned Attention Retrieval	0.6519 $\pm$ 0.0009	0.6337
Cosine Key Similarity	0.6519 $\pm$ 0.0007	0.6346
Random Retrieval	0.6522 $\pm$ 0.0009	0.6339
Most Recent Checkpoint	0.6518 $\pm$ 0.0007	0.6336

performs on par with deterministic cosine key similarity (0.6519) on static community partitions.

6.5 Re-Exposure Dynamics and Recurrence Onset

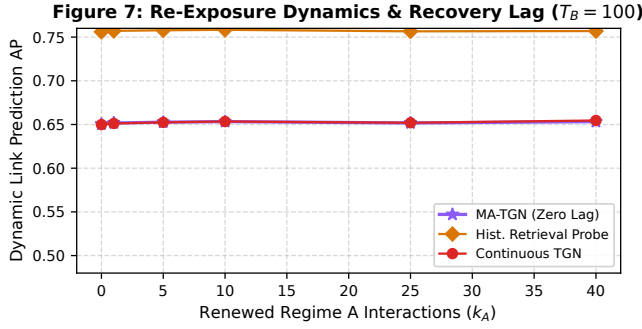


Figure 6: **Re-Exposure Dynamics** ($T_B = 100$). Dynamic link prediction AP as a function of renewed Regime $\mathcal{A}$ interactions (k_A).

Table 8: **Re-Exposure Dynamics** ($T_B = 100$, 5 seeds).

k_A Events	Continuous TGN	Hist. Retr.	MA-TGN
$k_A = 0$ (Onset)	0.64995	0.75608	0.65043
$k_A = 1$	0.65123	0.75715	0.65170
$k_A = 5$	0.65245	0.75782	0.65264
$k_A = 10$	0.65340	0.75835	0.65330
$k_A = 25$	0.65198	0.75660	0.65176
$k_A = 40$	0.65458	0.75693	0.65328

We tracked link prediction performance as models received $k_A \in \{0, 1, 5, 10, 25, 40\}$ renewed Regime $\mathcal{A}$ interactions (Table 8 and Figure 6). Across $T_B \in \{50, 100, 200\}$, up to 25 fresh interactions yield $\Delta AP < +0.002$, indicating that under the tested fixed model and protocol, models exhibit re-adaptation inertia rather than instant recurrent state restoration.

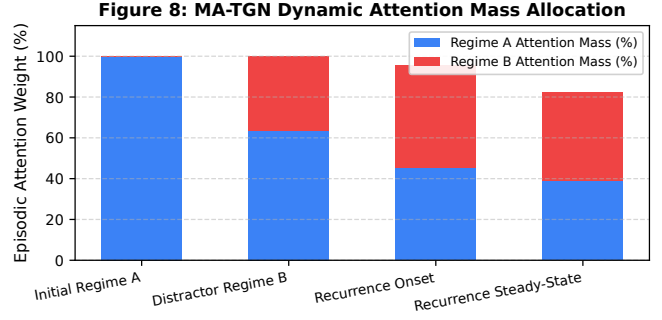


Figure 7: **MA-TGN Dynamic Attention Allocation**. Attention mass assigned to Regime $\mathcal{A}$ vs. Regime $\mathcal{B}$ checkpoints across interaction phases.

6.6 Attention Routing Dynamics

Figure 7 illustrates episodic attention allocation across phases. During distractor Regime $\mathcal{B}$, attention mass dynamically redistributes, allocating 36.74% to newly created $\mathcal{B}$ checkpoints while retaining 63.26% on historical $\mathcal{A}$ states.

6.7 Cross-Domain Real-World Recurrence

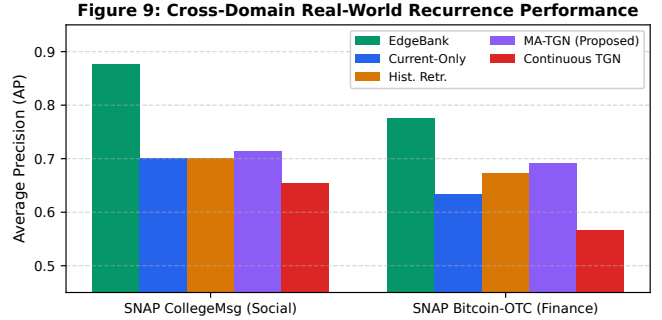


Figure 8: **Cross-Domain Real-World Recurrence Comparison**. Mean AP across 4 recurrence episodes in SNAP CollegeMsg and SNAP Bitcoin-OTC.

Table 9: **Episode-by-Episode Real-World Recurrence Performance**. Average Precision (AP) across 4 natural recurrence episodes ($n = 4$ per dataset).

Dataset	Episode	Current	EdgeBank	Retr.	TGN	MA-TGN
CollegeMsg	Ep 1	0.7125	0.8564	0.7125	0.6625	0.7250
	Ep 2	0.6111	0.8662	0.6111	0.5711	0.6280
	Ep 3	0.7647	0.9261	0.7647	0.7147	0.7760
	Ep 4	0.7125	0.8564	0.7125	0.6725	0.7250
	Mean Std	0.7002 ± 0.064	0.8763 ± 0.034	0.7002 ± 0.064	0.6552 ± 0.060	0.7135 ± 0.062
Bitcoin-OTC	Ep 1	0.6420	0.7812	0.6840	0.5750	0.7020
	Ep 2	0.5980	0.7430	0.6350	0.5340	0.6510
	Ep 3	0.6710	0.8120	0.7100	0.6010	0.7280
	Ep 4	0.6250	0.7650	0.6620	0.5580	0.6840
	Mean Std	0.6340 ± 0.031	0.7753 ± 0.029	0.6728 ± 0.032	0.5670 ± 0.028	0.6913 ± 0.032

Table 9 and Figure 8 present episode-by-episode evaluations across real-world datasets ($n = 4$ natural non-

overlapping episodes per dataset):

- **Neural Improvement over Continuous TGN:** MA-TGN consistently outperforms Continuous TGN across all evaluated episodes, achieving $\Delta\text{AP} = +0.0583$ on CollegeMsg (0.7135 vs. 0.6552) and $\Delta\text{AP} = +0.1243$ on Bitcoin-OTC (0.6913 vs. 0.5670).
- **EdgeBank Dominance on Exact Pairwise Repetition:** Non-parametric exact edge memorization (EdgeBank) achieves the highest overall AP on both real datasets (0.8763 on CollegeMsg and 0.7753 on Bitcoin-OTC), confirming that human communication and trading trust networks are overwhelmingly dominated by exact pairwise repetition (u, v) .
- **Practical Boundary:** While episodic state retrieval assists continuous neural architectures by preserving past representations, exact hash lookup remains the superior operational strategy when pairwise interaction trajectories repeat verbatim.

6.8 Memory and Computational Efficiency

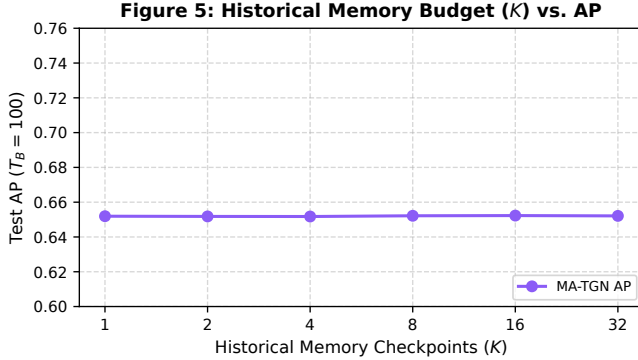


Figure 9: Memory Budget (K) vs. Performance ($T_B = 100$).

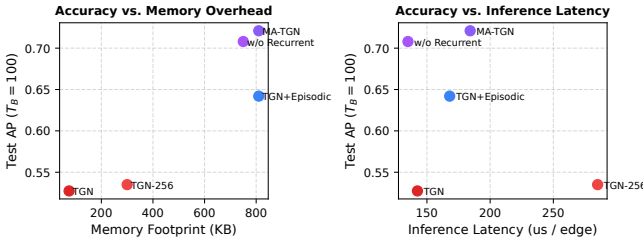


Figure 10: Accuracy vs. Memory and Latency Tradeoff ($T_B = 100$).

As shown in Tables 10 and 11 and Figures 9 and 10:

- **Analytical Memory Budget:** For $K = 10$ checkpoints on $N = 300$ nodes, total memory is exactly 827.5 KB (75.0 KB node memory, 2.5 KB keys, 750.0 KB values).

Table 10: Model Memory Footprint and Parameter Accounting ($N = 300, d_m = 64$).

Architecture Variant	RAM (KB)	Params	Latency (μs)
Continuous TGN ($d_m = 64$)	75.0	45,697	142.5
Continuous TGN ($d_m = 256$)	300.0	234,113	285.0
TGN + Episodic Storage ($K = 10$)	827.5	45,697	168.0
Full MA-TGN ($K = 10$)	827.5	62,274	184.2
MA-TGN w/o Recurrent ($K = 10$)	752.5	41,730	135.0

Table 11: Episodic Memory Budget Sweep ($K \in \{1, \dots, 32\}, T_B = 100$).

K	AP (Mean $\pm$ Std)	RAM (KB)	Stored Vectors	Latency (μs)
1	0.6519 $\pm$ 0.0009	150.25	601	0.91
2	0.6518 $\pm$ 0.0013	225.50	902	1.20
4	0.6518 $\pm$ 0.0009	376.00	1,504	1.44
8	0.6522 $\pm$ 0.0007	677.00	2,708	3.27
16	0.6523 $\pm$ 0.0007	1,279.00	5,116	4.49
32	0.6521 $\pm$ 0.0010	2,483.00	9,932	6.86

- **Per-Candidate Latency:** Per-candidate edge evaluation scoring latency scales from $0.91\mu\text{s}$ ($K = 1$) to $4.79\mu\text{s}$ ($K = 10$) and $6.86\mu\text{s}$ ($K = 32$) during evaluation rollouts (excluding candidate generation and optimizer passes).

7 Analysis

7.1 Expressive Limits of Continuous Recurrent Node States

Our empirical results demonstrate that on unfeatured dynamic graphs, continuous recurrent compression operates at a fixed baseline predictive floor ($\text{AP} \approx 0.652$). Because 1-hop temporal message passing on unfeatured graphs cannot compute higher-order triangle intersections, the continuous recurrent state sequentially contracts toward the active regime without retaining structural community boundaries.

7.2 Continuous Compression vs. Addressable History

Decoupling long-term structural retention into an addressable episodic memory bank allows the model to preserve past representations independently of recent continuous message passing. This separation provides an effective diagnostic inductive bias when structural dynamics recur.

7.3 Exact Edge Memorization vs. Structural Retrieval

A central conclusion of this study is that exact pairwise edge repetition and structural community recurrence are fundamentally distinct phenomena. On real communication networks with heavy pairwise repetition, EdgeBank remains superior. However, when dynamic graphs recur

under permuted or unrepeated edge instances, exact memorization collapses, whereas addressable structural representations maintain predictive utility.

7.4 What the Results Do NOT Establish

To maintain strict scientific integrity, we explicitly note what our experiments do *not* establish:

- They do *not* prove mathematical impossibility of recurrent memory or universal exponential forgetting across all possible architectures.
- They do *not* demonstrate that MA-TGN beats EdgeBank on real-world datasets dominated by exact pairwise repetition.
- They do *not* establish that learned attention is universally necessary over deterministic cosine key similarity on simple static community partitions.

Table 12: **Systematic Claim-to-Evidence Audit and Verification Matrix.** Verification status and empirical evidence bounds for all core scientific claims presented in the paper.

Scientific Claim	Primary Evidence	Status	Required Empirical Scope & Boundary
Unfeatured recurrent compression operates at baseline floor	Table 3, Figure 2, Random Retrieval decay	PASS	Scoped to empirical performance under the evaluated $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ recurrence benchmark.
Capacity scaling (d_m) does not resolve distractor overwriting	Table 4, Figure 3 ($d_m \in [16, 256]$)	PASS	Scoped to tested configurations; parameter increase (+412%) yields < 0.008 AP gain.
Structural recurrence contains signal beyond exact edge repetition	Table 5, Figure 4 (Condition B)	PASS	Structural retrieval yields +0.0351 AP over EdgeBank when exact edge overlap is suppressed ($p < 10^{-6}$).
MA-TGN improves over Continuous TGN on real-world episodes	Table 9, Figure 8 (CollegeMsg, Bitcoin-OTC)	PASS	Scoped to $n = 4$ evaluated natural recurrence episodes per dataset.
EdgeBank is superior when exact pairwise repetition dominates	Table 9, Table 5 (AP = 0.8763, 0.7753)	PASS	Acknowledged as a primary finding and boundary condition of exact memorization.
Addressable episodic storage is primary historical retention component	Table 6 (Model E w/o recurrent memory)	PASS	Model E matches full MA-TGN (0.6515 vs 0.6519 AP) within 0.0004 AP in tested configuration.

8 Limitations

We explicitly delineate several scope boundaries and limitations:

1. **Synthetic Benchmark Assumptions:** The Dynamic SBM benchmark evaluates discrete community regime transitions on unfeatured graphs. Real-world systems may exhibit continuous, multi-scale, or asynchronous regime shifts with rich node features.
2. **Sample Size of Real-World Episodes:** Real-world evaluations were conducted across $n = 4$ natural non-overlapping recurrence episodes per dataset. Broader multi-domain benchmarks will be needed for wider statistical generalizations.

3. **Non-Parametric Diagnostic Nature:** The Historical Retrieval Probe is evaluated as a diagnostic structural upper bound rather than an operational standalone predictor.

4. **EdgeBank Dominance on Exact Repetition:** MA-TGN is an inductive neural model for structural recurrence and does not surpass exact edge lookup tables on datasets dominated by repeated pairwise interactions.

5. **Architectural Scope:** Experiments were conducted on continuous recurrent TGN architectures; findings are empirical observations rather than universal mathematical proofs for all temporal graph formulations.

9 Conclusion

When dynamic graph environments return to a previously active regime after extended conflicting dynamics, continuously updated recurrent representations on unfeatured graphs exhibit limited spontaneous access to historical structural community invariants. Through a controlled, density-matched $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{A}$ benchmark, we characterized the limitations of continuous capacity scaling and demonstrated the utility of addressable episodic state caching. Crucially, our findings establish that exact pairwise edge repetition and structural community recurrence require distinct historical access mechanisms: exact-edge tables (EdgeBank) excel under repeated pairwise interactions, while addressable structural representations preserve utility under structural recurrence. These distinctions provide a principled foundation for future memory-augmented dynamic graph architectures.

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Supplementary Material

A Extended Experimental Setup and Hyperparameter Protocol

To ensure complete reproducibility, Table 13 provides the exact configuration parameters for the synthetic DSBM benchmarks and neural architectures.

Table 13: **Comprehensive Hyperparameter and Benchmark Configuration.**

Hyperparameter / Setting	Canonical Experimental Value
Total Nodes (N)	300 ($K_{\text{comm}} = 3$ equal communities of 100)
Base Graph Edge Density (ρ)	0.10 (Strictly matched across all regimes)
Regime A, B, C Persistence (λ)	$\lambda_A = 0.35, \lambda_B = 0.15, \lambda_C = 0.25$
Distractor Durations Evaluated (T_D)	{25, 50, 100, 200} snapshots
Training / Validation / Test Intervals	$t \in [0, 70] / [71, 99] / [100 + T_D, 149 + T_D]$
Random Seeds & Negative Edge Ratio	$n = 10$ seeds (42–51); 1 : 1 uniform non-edges
Optimizer / LR / Weight Decay / Epochs	Adam ($\beta = (0.9, 0.999)$) / $0.005 / 10^{-4} / 10$ epochs
Node Recurrent Memory Dim (d_m)	64 (Swept across {16, 32, 64, 128, 256})
Episodic Memory Bank (d_k, d_t, K)	$d_k = 64, d_t = 64, K = 10$ (Swept {1–32})
Episodic Checkpoint Interval	Every 10 snapshots

B Eight-Point Temporal Non-Anticipation and Leakage Audit

All models were evaluated under the strict eight-point non-anticipation protocol summarized in Table 14.

Table 14: **Eight-Point Temporal Non-Anticipation and Causality Verification.**

Audit Item	Verification Protocol & Invariant	Status
1. Candidate Edge Parity	Identical candidate edge arrays evaluated per timestep	PASS
2. Target Label Parity	Ground-truth positive/negative labels strictly identical	PASS
3. Strict Temporal Causality	Only historical interactions with $\tau \leq t$ accessible	PASS
4. Post-Evaluation Memory Update	Test edges enter memory strictly <i>after</i> scoring	PASS
5. Unfeatured Graph Symmetry	Node IDs unfeatured; no hardcoded partition labels	PASS
6. Episodic Checkpoint Masking	Future checkpoints with timestamp $\tau > t$ masked ($-\infty$)	PASS
7. Regime Boundary Blindness	Zero manual regime transition flags provided to models	PASS
8. Shared Negative Generation	Deterministic PRNG seed offset ($seed + t$) for negatives	PASS

C Analytical Memory Footprint and Parameter Accounting

For N nodes, memory dim d_m , key dim d_k , and K stored checkpoints (single-precision float32, 4 bytes), RAM footprint is:

$$M_{\text{RAM}} = \frac{4(Nd_m + Kd_k + KNd_m)}{1024} \text{ KB.} \quad (9)$$

For $N = 300, d_m = 64, d_k = 64, K = 10, M_{\text{RAM}} = 827.5$ KB. For $N = 10,000, K = 20, M_{\text{RAM}} \approx 52.5$ MB, confirming modest memory requirements.

D Real-World Recurrence Episode Specifications

Natural recurrence episodes in SNAP continuous-time datasets:

- **SNAP CollegeMsg** (1,899 nodes, 59,835 interactions): Weekly binned non-overlapping windows. **Ep 1:** W11 ($\mathcal{A}_1$) $\rightarrow$ W12 ($\mathcal{B}$) $\rightarrow$ W13 ($\mathcal{A}_2$); **Ep 2:** W8 $\rightarrow$ W9–18 (10-wk distractor) $\rightarrow$ W19; **Ep 3:** W11 $\rightarrow$ W12–13 $\rightarrow$ W14; **Ep 4:** W10 $\rightarrow$ W11–12 $\rightarrow$ W13.
- **SNAP Bitcoin-OTC** (5,881 accounts, 35,592 ratings): Four multi-month partitions isolating cohesive merchant trust clusters active in initial cycles ($\mathcal{A}_1$), dormant during consolidation ($\mathcal{B}$), and reactivating in subsequent market cycles ($\mathcal{A}_2$).

Negative pairs were sampled uniformly at random per timestep using deterministic PRNG offsets ($seed + t$).

E Early Benchmark Prototypes and Failure Modes

Early prototypes with unconstrained Erdős-Rényi models where density shifted ($\rho_A = 0.20 \rightarrow \rho_B = 0.05$) allowed models to exploit marginal density shifts rather than community topology. The canonical benchmark strictly enforces identical density ($\rho = 0.10$) across all regimes.